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Rushmore Consultation

Team C

ACS 5513 – Machine Learning Practice
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Project Charter

Business background

- Our clients are voter outreach organizations that want to drive civic engagement. These organizations often need to decide which people should receive reminder calls, texts, emails, or educational materials before an election. Blanket outreach is expensive and inefficient, while manual prioritization tends to rely on intuition rather than evidence.
- These organizations want to identify people with a high risk of not voting so outreach can be prioritized. Using the FiveThirtyEight non-voters survey data, the team will build a machine learning application that predicts a respondent's voting behavior category from demographic and attitudinal variables

Business proposition

This project's core proposition is that civic outreach groups can spend limited outreach dollars more intelligently by prioritizing the people who are least likely to vote without intervention. Instead of treating every contact the same, the organization can use a modeled turnout-risk score to segment respondents into action tiers and reserve the most expensive outreach channels, such as canvassing and live calls, for the people who need them most.

The primary users are nonprofit voter engagement teams, advocacy groups, and campaign operations staff who already run outreach programs but lack a consistent data-driven way to decide who should be contacted first. For these users, the product reduces guesswork, makes targeting decisions easier to explain to stakeholders, and creates a repeatable process that can be reused from one election cycle to the next.

The business value comes from improving both efficiency and measurable impact. If the model identifies low-propensity voters more accurately than blanket outreach or intuition-based lists, the organization can lower cost per meaningful contact, improve volunteer time allocation, and increase turnout lift in the populations it serves. Even a modest improvement in response targeting can translate into better use of campaign budgets, stronger reporting to donors or leadership, and a clearer justification for future outreach investment.

From an operational perspective, the delivered solution is more than a one-time prediction file. The team intends to provide a reproducible Python pipeline that can ingest updated survey-style data, retrain the model, and produce interpretable predictions with confidence levels and feature explanations. That makes the solution realistic for ongoing use: the client receives not only a ranked list of outreach candidates, but also insight into why a respondent is being flagged and which communication method may be the most appropriate next step.

Scope

- We will build a Python-based classification pipeline that predicts voter turnout propensity and surfaces the most important drivers behind each prediction.
- How we will accomplish this
    - Acquire and profile the FiveThirtyEight non-voters dataset.
    - Clean missing values and encode categorical survey responses.
    - Engineer a reproducible training dataset for multiclass classification.
    - Train and compare baseline and improved models using scikit-learn.
- The customer will receive our model's predictions, recommended avenues for voter registration, door-knocking, flyers, and other methods of driving voter turnout

Personnel

- Andrew Lee
- Jakobi Oommen
- Sisi Malone

Metrics

- The purpose of this project is to provide a usable and explainable model for identifying turnout-risk respondents.
- The clients will see success when they can meaningfully increase voter turnout in a district by contacting the prospects we provide them.
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- The baseline metric is whatever increase in voter turnout the organization has managed to achieve without our model. Be that +20% voter turnout, we'd like to see that metric increase to +25% to +30% in a real world example.
- This can be measured using voting data published by the government post-election.

Architecture

Expected data: one CSV survey dataset from FiveThirtyEight containing 5,836 respondent records and 119 columns. The source includes survey response codes, demographic fields, a survey weight, and the turnout label.

Python solution stack: pandas for data ingestion and cleaning, numpy for transformations, scikit-learn for preprocessing pipelines and model training, matplotlib/seaborn for visualization, joblib for model serialization.

End-to-end flow:

Raw CSV from GitHub -> Python ingestion script -> cleaned training dataset -> preprocessing pipeline (imputation, encoding, scaling if needed) -> classification model -> saved model artifact -> prediction, confidence, and feature explanation for end user.

Customer consumption: a user enters or uploads respondent-style features, the app returns predicted turnout category and confidence, then highlights which factors drove the score.

Planned production-style pipeline: a retraining script will rerun on updated survey data, regenerate features, train the model, save the best artifact, and refresh the deployed app's model package.

Communication

- Weekly team meeting for planning, progress updates, and issue resolution.
- Shared Git repository with feature branches and pull requests for code review.
- Shared task board documenting ownership, due dates, and iteration goals.
- Checkpoint review video recorded collaboratively with each member presenting their assigned section.

Data Report

Data Sources

Raw Data Sources

Dataset Name	Original Location	Destination Location	Data Movement Tools / Scripts	Link to Report
FiveThirtyEight Non-Voters Survey Data	GitHub raw CSV hosted by FiveThirtyEight: `https://raw.githubusercontent.com/fivethirtyeight/data/master/non-voters/nonvoters_data.csv`	Local project data folder, e.g. `data/raw/nonvoters_data.csv`	`download_data.py` or direct pandas read from URL	This Data Report

The project uses the FiveThirtyEight non-voters survey dataset as the primary and sufficient data source. The data can be accessed directly from GitHub and parsed into Python using pandas. The dataset contains demographics, political attitudes, civic perceptions, issue-based questions, and a labeled turnout behavior category. We constructed a script called `parse_data.py` to conduct an exploratory data analysis to better understand the scope of the data we intended to work with. This script reports the shape of the data set, analyses the datatypes, and investigates which columns have sufficient missing values that need cleaning. The script also helped us discover that we have a clear usable 3-class target variable, `voter_category`, that doesn't suffer from a balance issue.

The cleaning script, `clean_data.py`, begins by correcting a labeling error in the raw survey data: the value -1 does not mean the same thing across all columns. For 28 "select all that apply" columns (the Q19, Q28, and Q29 batteries), -1 and 1 are the only values present, meaning -1 represents a real, informative "not selected" response rather than a missing one; these are recoded to a clean 0/1 binary indicator. For the remaining 83 true ordinal columns, -1 genuinely represents a refused or skipped response and is recoded to NaN. After this correction, 4,071 instances of true refusal are converted to NaN across the scale columns, and any column with more than 50% missing data is dropped as unreliable. Remaining missingness (8,037 values across 97 survey columns) is filled using each column's median, which avoids introducing fractional responses that were never valid answer choices. The four demographic fields are then encoded for modeling: educ and income_cat, which have a natural low-to-high ordering, are mapped to ordinal integers, while race and gender, which have no inherent order, are one-hot encoded to avoid implying a false ranking between categories. Finally, the script builds a voting_obstacles_composite feature by averaging the ten Q18 sub-items into a single interpretable score, reducing redundancy from near-duplicate columns. The result is a cleaned dataset of 5,836 respondents and 112 columns, with the three-class target distribution (44.1% sporadic, 31.0% always, 24.9% rarely/never) preserved exactly from the raw data, confirming the cleaning process didn't distort the underlying class balance.

Data Dictionary

The table below summarizes the fields and field groups used in the FiveThirtyEight non-voters dataset. The Q-coded variables follow the source survey codebook, while the demographic and target fields provide the main business-facing context for the classification task.

Field or Group	Role / Type	Business Meaning	Modeling Notes
RespId	Identifier	Unique respondent ID from the survey export.	Used for traceability only; excluded from model training to avoid meaningless signal.
weight	Numeric weight	Survey weight for producing representative descriptive statistics.	Useful for reporting and weighted analysis; optional in baseline model training.
Q1	Ordinal survey response	General civic or voting-related	Retained only after codebook review

		survey response captured by the source instrument.	and missing-value normalization.
Q2_1 - Q2_10	Binary / multi-select survey items	Battery of related responses recorded as separate columns.	Some of these are true checked / not checked flags and require careful treatment of -1 values.
Q3_1 - Q3_6	Binary / multi-select survey items	Follow-up attitude or behavior battery represented across six columns.	Likely encoded for one-hot style analysis after cleaning.
Q4_1 - Q4_6	Binary / multi-select survey items	Issue or perception battery captured as separate indicator fields.	Candidate feature group for attitudes linked to turnout behavior.
Q5 - Q7	Ordinal survey responses	Standalone survey questions with coded numeric answers.	Will be imputed or encoded according to the codebook-defined response scales.
Q8_1 - Q8_9	Binary / multi-select survey items	Another family of related survey prompts decomposed into columns.	Can be kept as indicator features after recoding nonresponse correctly.
Q9_1 - Q10_4	Binary / ordinal survey items	Grouped topical responses about attitudes, behaviors, or priorities.	Reviewed for sparsity and predictive value before inclusion.
Q11_1 - Q11_6	Binary / ordinal survey items	Issue battery captured across six related columns.	May contribute interpretable issue-salience signals.
Q14 - Q16	Ordinal survey responses	Mid-survey standalone coded responses.	Handled as categorical or ordinal depending on codebook semantics.
Q17_1 - Q19_10	Binary / ordinal survey batteries	Large family of civic, issue, or institutional trust responses.	High-dimensional block that may be useful in the richer model variant.

Q20 - Q26	Ordinal survey responses	Later survey questions represented as single coded columns.	Potentially strong turnout predictors if missingness remains acceptable.
Q27_1 - Q28_8	Binary / multi-select survey batteries	Grouped responses stored as one column per option.	Need the same binary-flag cleaning logic identified during EDA.
Q29_1 - Q29_10	Sparse survey battery	Late-stage survey item family with substantial missingness.	Likely candidate for exclusion, grouped imputation, or sensitivity testing.
Q30 - Q33	Ordinal survey responses	Final standalone coded responses in the instrument.	Included only if they do not leak the target and pass quality checks.
ppage	Numeric demographic	Respondent age in years.	High-value baseline predictor and useful segmentation variable for outreach planning.
educ	Categorical demographic	Education attainment category.	Expected to be one-hot encoded in the preprocessing pipeline.
race	Categorical demographic	Respondent race category.	Used for modeling and fairness-aware interpretation.
gender	Categorical demographic	Respondent gender category.	Used as a baseline feature and for subgroup reporting.
income_cat	Categorical demographic	Household income band.	Likely predictive of turnout propensity and useful for segment analysis.
voter_category	Target label	Observed turnout frequency class: always, sporadic, or rarely/never.	Prediction target for the multiclass classification model.

Deliverable Video

Github